

SEMESTER - S3/S4			
ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT (Common to all Branches)			
Course Code	UCHUT347	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	40
Credits	2	Exam Hours	-
Prerequisites (if any)	None/ (Course code)	Course Type	Theory
Course objectives: - Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives. - Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development. - Develop the ability to find strategies for implementing sustainable engineering solutions.			
Syllabus			Contact Hours
Module-I	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism , Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution -Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places -accessibility and social impacts, Managing conflict , Collective bargaining, Confidentiality , Role of confidentiality in moral integrity, Codes of Ethics . Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education , employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives .		6hrs
Module-II	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.		6 hrs
Module- III	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste		6 hrs

	initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	
Module- IV	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	6 hrs

Course Assessment Method (CIE: -100 Marks, ESE: 0 Marks)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through the HMC courses and Miniproject course (S6).

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	6
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	6
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	6
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	18
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	9
Total Marks				60

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Part A	Part B	Total
Four short answer questions with five marks each, one from each module. (6x3 = 18 marks)	Part B contains 2 questions from each module of which student should answer any one. (4x8 = 32 marks)	50 marks

Course Outcomes (COs)		
At the end of the course the student will be able to:		
Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3
K1- Remember, K2- Understand, K3- Apply, K4- Analyze, K5- Evaluate, K6- Create		

Course Articulation Matrix (Mapping of course outcomes with program outcomes):												
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2
1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation												

Reference books				
SL No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S	PHI Learning Private Ltd,	2012

		Natarajan and V S Senthil Kumar	New Delhi	
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014
8	Engineering Ethics,	Charles D Fleddermann,	Pearson Education/ Prentice Hall of India, New Jersey	2004.
9	Engineering Ethics- Concepts and cases	Charles E Harris, Michael S Protchard and Michael J Rabins	Wadsworth Thompson Learning, United states,	2005.
10	Sustainability Engineering: Concepts, Design and Case Studies,	Allen, D. T. and Shonnard, D. R.	Prentice Hall	2011
11	Delusions of Gender: How Our Minds, Society, and Neurosexism Create Difference	Fine, Cordelia	W.W. Norton & Company	2011

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.

- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT347				
Course Name: ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT				
Max. Marks: 50			Duration: 2 hours 30 minutes	
PART A				
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What is civic virtue? How does it relate to professional ethics?	1	(3)
2		What do you mean by 'gendered technology' and provide one specific example of a product whose design or application reflects this concept.	2	(3)
3		Explain the term triple bottom line in sustainable engineering.	3	(3)
4		Why is biodiversity considered essential for ecosystem stability? Give one example.	3	(3)
5		What is the idea behind degrowth? How is it different from economic development?	4	(3)
6		Identify two engineering solutions that can reduce greenhouse gas emissions in everyday life.	5	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9		You are part of a team designing an app that collects user data for public services. What ethical concerns would you raise regarding data privacy and consent? Suggest safeguards.	1	(8)
10	a)	What is the fundamental difference between 'Equity' and 'Equality' when discussing gender initiatives and resource allocation in education/workplace?	2	(5)
	b)	Suggest measures to promote gender equity in workplaces.	2	(3)
Module 2				
11		A new industrial project near a wetland promises jobs but threatens local ecosystems. Apply the three ethical lenses (anthropocentric, biocentric, ecocentric) to assess it and recommend a balanced decision.	3	(8)
12	a)	Explain how life-cycle analysis (LCA) can help engineers design more sustainable products.	3	(5)
	b)	Illustrate with one example relevant to Kerala.	3	(3)

Module 3				
13		Rapid motorization is worsening air pollution. Suggest sustainable mobility options suitable for medium-sized Kerala towns, explaining their environmental benefits.	4	(8)
14		Outline a three-step plan to move a college campus towards zero waste. Mention the role of engineers, students, and the college management in each step.	4	(8)
Module 4				
15	a)	Discuss how renewable energy technologies (solar, wind, biogas) can be scaled up in Kerala.	5	(4)
	b)	State two technical and two social challenges to their adoption.	5	(4)
16	a)	Discuss the importance of environmental policies and regulations in promoting sustainable engineering.	5	(4)
	b)	Give one national and one international example and their impact.	5	(4)
